Tutorial 5: Linear Transformations

22 October 2025

1. Find all matrices $\boldsymbol{X}$ that satisfy each of the following matrix equations.

   (a) $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \boldsymbol{X} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

   (b) $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix} \boldsymbol{X} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

   (c) $\begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix} \boldsymbol{X} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

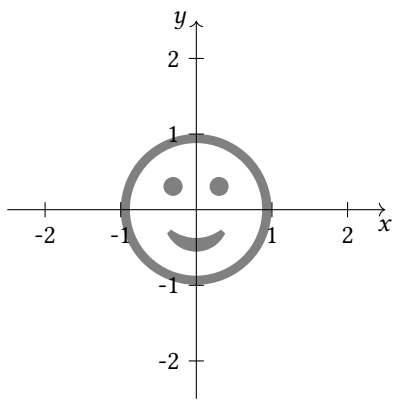
2. Let the 2×2 matrices $\boldsymbol{A}$ be given by

$$\boldsymbol{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

   Show that if $ad - bc \neq 0$, then $\boldsymbol{A}$ is invertible and

$$\boldsymbol{A}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

3. Consider the following image on $\mathbb{R}^2$.



   For each of the following matrices $\boldsymbol{A}$, draw a sketch showing the effect the transformation $T(\vec{x}) = \boldsymbol{A}\vec{x}$ on the face.

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

4. Let $T(\vec{x}) = \text{proj}_\ell(\vec{x})$ be the transformation corresponding to the projection of $\vec{x} \in \mathbb{R}^2$ onto a line ℓ passing through the origin.

   (a) What is $T(T(\vec{x}))$? Justify your answer.

   (b) Is $T(\vec{x})$ invertible? Justify your answer.

5. Let ℓ be a line in $\mathbb{R}^2$ passing through the origin, parallel to the non-zero vector $\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$. Let $T(\vec{x}) = \mathrm{ref}_\ell(\vec{x})$ be the transformation corresponding to the reflection of $\vec{x} \in \mathbb{R}^2$ about the line ℓ, and let

$$\boldsymbol{A} = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}, \text{ where } a^2 + b^2 = 1.$$

   (a) Show that the standard matrix of $T(\vec{x})$ is given by a matrix of the form $\boldsymbol{A}$.

   (b) Conversely, show that $\boldsymbol{A}$ must represent a reflection about some line. [*Hint:* Find non-zero, orthogonal vectors $\vec{v}, \vec{w}$ such that $\boldsymbol{A}\vec{v} = \vec{v}$ and $\boldsymbol{A}\vec{w} = -\vec{w}$, and show that $\boldsymbol{A}\vec{x}$ reflects $\vec{x}$ about the line $\ell = t\vec{v}$.]

   (c) Determine the inverse of the matrix $\boldsymbol{A}$.

6. The standard matrix of a rotation by a fixed angle α in the counter-clockwise direction is given by

$$\boldsymbol{R}_\alpha = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}.$$

   (a) Let $\vec{v} = \begin{pmatrix} \cos\beta \\ \sin\beta \end{pmatrix}$. Geometrically, explain why

$$\boldsymbol{R}_\alpha\vec{v} = \begin{pmatrix} \cos(\alpha+\beta) \\ \sin(\alpha+\beta) \end{pmatrix}.$$

   Hence, show that

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

   (b) Compute the matrix products $\boldsymbol{R}_\alpha\boldsymbol{R}_\beta$ and $\boldsymbol{R}_\beta\boldsymbol{R}_\alpha$ for some angles α, β. Give a geometric description of the transformation that these products represent.

   (c) Determine the inverse of $\boldsymbol{R}_\theta$, and give a geometric description of the linear transformation with standard matrix $\boldsymbol{R}_\theta^{-1}$.

7. For each of the following sequences of transformations in $\mathbb{R}^2$: (i) find the standard matrix of the composite transformation, and (ii) find the standard matrix of the composite transformation when the transformations are performed in the *reverse* order.

   (a) A rotation by $90°$, followed by a reflection about the line $y = x$.

   (b) A scaling by a factor of 2, followed by a rotation by $45°$, followed by a reflection about the y-axis.

8. Let $\boldsymbol{A}$ be an $n \times n$ matrix.

   (a) Show that if $\boldsymbol{A}^2 = \boldsymbol{0}$, then $\boldsymbol{I} - \boldsymbol{A}$ is invertible and $(\boldsymbol{I} - \boldsymbol{A})^{-1} = \boldsymbol{I} + \boldsymbol{A}$.

   (b) Show that if $\boldsymbol{A}^3 = \boldsymbol{0}$, then $\boldsymbol{I} - \boldsymbol{A}$ is invertible and

$$(\boldsymbol{I} - \boldsymbol{A})^{-1} = \boldsymbol{I} + \boldsymbol{A} + \boldsymbol{A}^2.$$

   (c) If $\boldsymbol{A}^n = \boldsymbol{0}$ for some $n \geq 4$, is $\boldsymbol{I} - \boldsymbol{A}$ invertible?